INSTITUTIONAL INVESTMENT ANALYTICS

Insider Alpha

Transforming SEC Form 4 Filings into
High-Conviction Investment Signals

For Portfolio Managers & Institutional Investors

THE PROBLEM

40,000+

Form 4 filings
annually in the S&P 500

Information Asymmetry

Corporate insiders legally must disclose trades — yet few investors systematically exploit this data.

Manual Analysis Impossible

Thousands of daily filings make it impractical to track, normalize, and extract signals by hand.

Missed Alpha Opportunities

Legal insider activity has proven to predict short-to-medium term price moves — a signal being left on the table.

The S&P 500 is the world's most efficient market — yet this information gap persists.

THE SOLUTION

Insider Alpha consolidates and distills SEC Form 4 data for S&P 500 companies into a unified, signal-ready analytics dashboard.

Hierarchical Weighting

Prioritizes C-suite trades (CEO, CFO) over lower-level employee transactions to surface the highest-conviction signals.

Cluster Detection

Flags 'Isolated/Significant Purchases' — instances where multiple independent insiders buy simultaneously.

Temporal Analytics

Time-stamps each signal to capture alpha within the critical 1–6 month post-filing window.

Algorithm automatically excludes non-discretionary trades: stock grants, option exercises & 10b5-1 pre-planned sales.

PILLAR 1

Value: Quantifiable Alpha Generation

5–10%

average outperformance
in the 6 months following
cluster insider purchases

📚 **Sources**

Jeng, Metrick & Zeckhauser (2003)
Estimating the Returns to Insider Trading
jstor.org/stable/3211592

Alldredge & Blank (2017)
Do Insiders Cluster Trades with Colleagues?
papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID3299023_code2385328.pdf?abstractid=2781761&mirid=1

Systematic Scoring

Aggregates data into a composite score. PMs set strict risk thresholds — execute only when criteria are met.

Risk Mitigation

Isolates CEO sell signals as early warnings for downside risk — enabling hedging or exit before negative surprises.

Uncorrelated Alpha

Returns uncorrelated with traditional factor models (value, momentum). A distinct and differentiated edge.

Research-backed: Academic studies confirm the predictive power of insider cluster purchases across market cycles.

PILLAR 2

Efficiency: Streamlining Operational Workflows

Noise Filtering

- Automatically excludes non-discretionary trades
- Removes stock grants, tax-related sales & option exercises
- Eliminates false positives — only opportunistic trades where insiders risk their own capital
- Algorithm grounded in academic research methodology

Unified Dashboard

- Replaces archaic SEC EDGAR navigation
- Normalized, sortable views across all S&P 500 sectors
- Buying/selling trend visualization at a glance
- Reduces hours of manual research to seconds

Insider Alpha transforms a 40,000+ filing annual workload into a streamlined, decision-ready workflow.

PILLAR 3

Strategic Alignment: Data-Driven Decision Making

Thematic Analysis

Track which sectors (Tech, Healthcare, Financials) show the highest insider buying concentration — enabling top-down allocation with bottom-up conviction.

Compliance Ready

Built entirely on public, auditable SEC filings. Clear audit trail for compliance departments on the origin of every trade idea.

Proprietary Data Moat

Transforms legal insider filings into a systematic process. A source of return not universally available to passive market participants.

Benchmark Outperformance

Operationalizing insider data enables asset managers to capture alpha uncorrelated with traditional factor-based strategies.

CONCLUSION

Your Edge in the S&P 500 Landscape

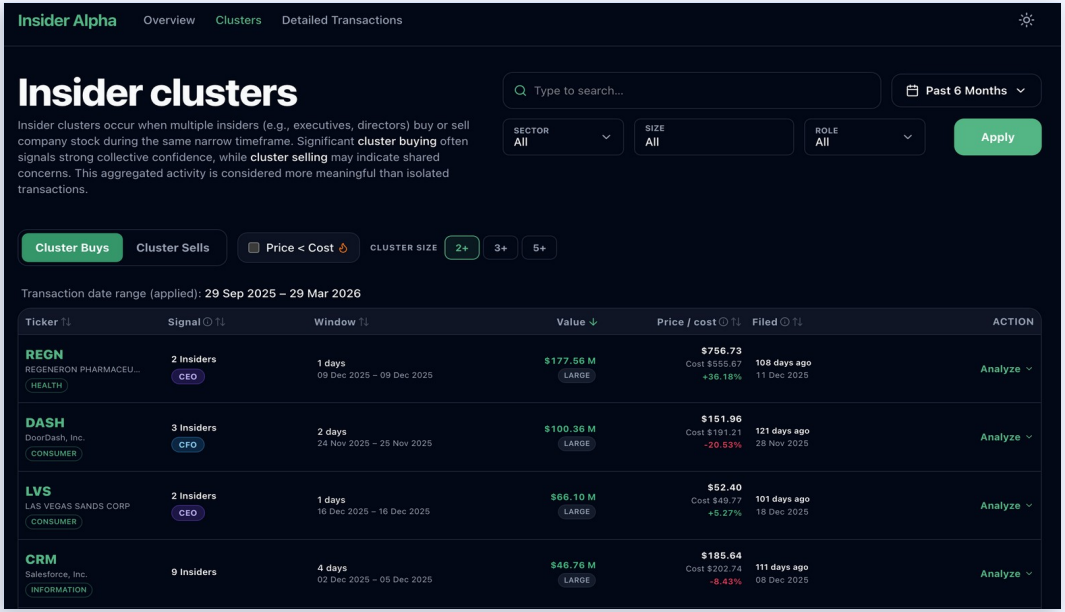
Insider Alpha bridges the gap between public regulatory data and proprietary investment insight — applying a proven methodology to the 'smart money' trades of C-suite executives.

1–6 mo.	5–10%	40,000+
Alpha capture window	Benchmark outperformance	Annual filings processed

High-conviction · Actionable · Compliant · Data-driven

LIVE PRODUCT

See It in Action



Insider Clusters Dashboard

REGN Stock Chart with Transaction Date

Scan to Try It Live

insideralpha.theluwak.com

Live demo available now

UNDER THE HOOD

Technical Solution Overview

Technical Architecture: ELT Pipeline for Insider Trading Analytics

Data Ingestion & Extraction (EL)

SEC EDGAR
Form 4: ~1kl+ annual records

DataHub
S&P 500 constituent liste

Yahoo Finance API
Daily price history

Automated Wrapper Scripts
Python & Shell Scripts fetch OHLCV data in JSONL

Meltano (Ingestion)
Multi-Source Raw Data Extraction

JSONL
Optimized Clustering on Primary Keys for Immediate Ingestion

Staging Tables
High-Efficiency Loading
Batch & Incremental Updates

Google BigQuery

Pipeline Orchestration (Dagster)

Dagster

Asset-Based Orchestration
Coordinate "sec_direct_ingestion" assets "dbt_insider_transformation"

Automated Scheduling Cycles
Quarterly (Foll SEC) — Monthly (Schema Validation) — Weekly (General Health Checks)

Observability & Error Handling

Data Quality & Transformation (dbt)

100% Critical Field Coverage
Unique on primary keys
Not Null on primary keys

dbt

Financial Business Logic Validation

```
custom macros non_negative_or_null (
  flag nsence >> (
    'non_negative_or_null' flagg errors
    in share count or prices
end
```

Data Cleaning & Standardization

Normalize Filing dates
- March 16, 2024
- March 15, 2023
- March 15, 2022

Standardize Ucber cymbols & transaction codes
- P -> Purchase
- S -> Sale

Data Warehouse Design (Star Schema)

dim_sec_submission
Grain: ACCESSION_NUMBER
Deduplication, Date Parsing

dim_sec_reporting_owner
Grain: Accession + Owner
Role Classification

fct_sec_nonderiv_line
Grain: Accession + Line SK
Share x Price Value Calculations

Fact Table
fct_insider_transactions
Grain: ACCESSION_NUMBER
Aggregates filing level data, calculates derived metrics

dim_sp500_company
Grain: symbol_norm
CIK-to-Ticker Mapping, GICS Sector

Performance-Optimized Serving Layer
sp500_insider_transactions
Year Partitioning on TRANS_DATE
Multi-Column Clustering on symbols
Fast API Response Times

Visualization & Dashboard
FcstAPI
NEXT.JS
Real time access in the dashboard

Key Star Schema Components

Layer	Table Name	Grain / Primary Key	Key Logio
Dimensions	dim_sec_submission	ACCESSION_NUMBER	Deduplication, Date Parsing
	dim_sec_reporting_owner	Accession + Owner	Role (Director/Officer) Classification
	dim_sp500_company	symbol_norm	Cltl.to-Ticker Mapping, GICS Sector
Facts	fct_sec_nonderiv_line	Accession + Line SK	Share x Price Value Calculations
	fct_insider_transactions	ACCESSION_NUMBER	Rolling up transactions to filing level
Serving	sp500_insider_transactions	ACCESSION_NUMBER	Year Partitioning, S&P 500 Filtering

Q&A

Questions & Answers

We'd love to hear from you.

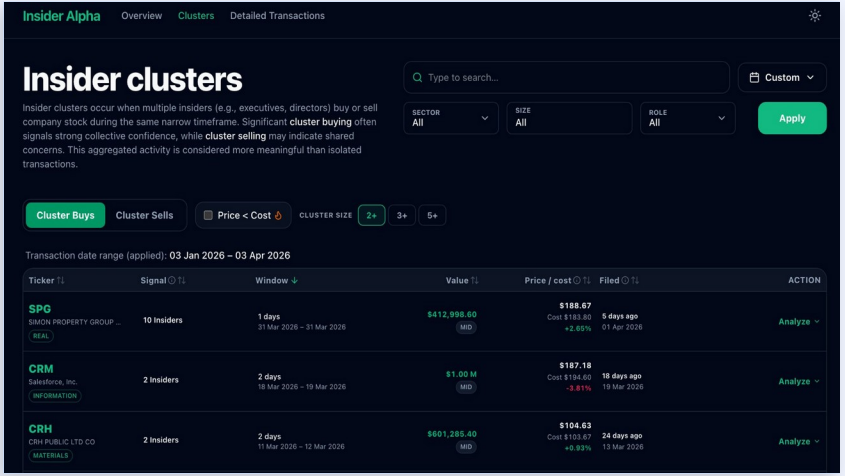
insideralpha.theluwak.com

Key Academic Research & Market Analysis

Pattern	Academic Sources	Market Analysis
Following the Leaders CEO/CFO trades carry the most predictive weight	Knewtson & Nofsinger (2014) — Why Are CFO Insider Trades More Informative? Wang, Shin & Francis (2012) — Are CFOs' Trades More Informative than CEOs'? Sabherwal & Uddin (2019) — Star CEOs and Insider Trading	GuruFocus: Insider Trading Strategy Seeking Alpha: CFO Insider Trading Anomaly
Cluster Purchases Multiple insiders buying together signals strong conviction	Alldredge & Blank (2017) — Do Insiders Cluster Trades with Colleagues? Lim & Park (2018) — Insider Herding and Stock Price Performance	Buyback Analytics: Cluster Buying Methodology
1–6 Month Window Alpha is strongest immediately after the filing date	Jeng, Metrick & Zeckhauser (2003) — Estimating Returns to Insider Trading Swedish Study (2014) — Abnormal Returns from Insider Trading Sabherwal & Uddin (2019) — 6-month horizon	SmartInsiderTrades: Methodology GuruFocus: Short-Term Market Direction Seeking Alpha: CFO Anomaly Timing

Sources: SSRN, JSTOR, DIVA Portal, GuruFocus, Seeking Alpha, BuybackAnalytics, SmartInsiderTrades

How to Capture Alpha: A Step-by-Step Use Case



Live dashboard — insideralpha.theluwak.com

SPG

10 insiders · 1-day window · strongest cluster signal

CRM

$1M from 2 insiders · price -3.81% below insider cost

1 Identify the Cluster Signal

Look for multiple insiders buying in a narrow window. SPG: 10 insiders in 1 day = strong collective confidence.

2 Assess Conviction (Value)

Size matters. CRM: $1M from 2 insiders signals real skin in the game. Minimum threshold: $100k+.

3 Check the Filing Lag

Insiders file Form 4 within 2 business days. Prioritize 'fresh' clusters — before the market prices in the news.

4 Price vs. Cost Gap

Red % = stock trades below insider entry. CRM at -3.81% means you can enter at a better price than the CFO.

5 Fundamental Contextualization

Use Analyze to confirm: bottom-fisher or inflection buy? Does this insider have a strong historical track record?